AI-Driven Detection of Iron Deficiency: Harnessing Visual Cues for Early Diagnosis and Anemia Prevention

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ABSTRACT

Iron deficiency is one of the most common and widespread micronutrient deficiencies, leading to conditions like anemia, fatigue, and cognitive impairments. Despite its prevalence, the subtlety of symptoms and challenges in accessing regular diagnostic testing often result in delayed detection and treatment. Traditional methods of diagnosing iron deficiency typically involve blood tests, which can be inaccessible, costly, and time-consuming. To address this issue, we propose an innovative approach using a Convolutional Neural Network (CNN) driven analysis of visual cues to detect iron deficiencies early and efficiently. By leveraging machine learning techniques, this method will assess biomarkers such as nail health, eye appearance, and tongue discoloration, which may correlate with iron deficiency. Through non-invasive data collection, this AI model will aim to provide a rapid, accessible, and cost-effective diagnostic tool, made easily available via a mobile app. The potential to revolutionize early and accessible detection could enable timely intervention and help reduce the global health burden of undiagnosed iron deficiencies.

MOTIVATION AND APPROACH

Iron deficiency affects over 25% of the global population and often progresses to anemia, a silent but deadly condition impacting half a billion women and 269 million children globally (IHME, 2023; WHO, 2023). Diagnosing iron deficiency relies on invasive CBC tests, which are resource-intensive, inaccessible, and time consuming (1- 4 hours in even resource rich settings), delaying early detection and intervention especially for those in low-resource communities

(Suner et al., 2021; McLean et al., 2005). Visible markers like pale conjunctiva, brittle nails, and glossitis offer diagnostic potential (The Iron Clinic, 2024).

Currently, AI driven apps to detect micronutrient deficiencies, particularly iron deficiency, are rare and are not multimodal. Our work is similar to that of (Nandi et al., 2018) and (Zhao et al., 2024), whose research to detect anemia works by analyzing the conjunctiva (inner surface of the under eyelid) to predict hemoglobin levels. However, as both studies are somewhat similar and work through a unimodal approach of analyzing images of just the patient's conjunctiva, the accuracy of this approach for detecting hemoglobin levels was ~75% (Zhao et al, 2024). Other, more recent work, such as that by (Ramzan et al., 2024), has explored a multimodal approach by integrating conjunctival image analysis with data from patients' Electronic Health Records (EHRs) to achieve higher diagnostic accuracy. While this approach demonstrates the potential of multimodal techniques (with a 95% accuracy), its reliance on EHRs poses significant limitations for immediate, fast diagnosis and accessibility, especially in rural or economically disadvantaged populations where EHRs may not be readily available for immediate, early diagnosis.

Building on these findings, we aim to develop a truly accessible and scalable multimodal solution that combines analyses of the patient's hemoglobin levels through conjunctival analysis, along with nail health, and tongue and skin discoloration—three common, non-invasive, and early indicators of iron deficiency—using only a smartphone app and camera. Unlike previous apps/technologies that are in play, our goal isn't to detect anemia, but to address the root cause and detect iron deficiencies, in order for users to take preventative actions before the development of anemia. To achieve accurate and reliable predictions, our system will employ a convolutional neural network (CNN) for feature extraction from images, also using a Residual

Convolutional Block Attention Module (RCBAM) in the preliminary layers and a Dual-Head Attention Network (DHANet) in the later layers. The RCBAM focuses on refining both spatial and channel attention in the earlier layers of the network, helping to quickly identify and emphasize critical features (Zubair, 2024) in the input images, such as the vascular patterns of the conjunctiva or texture anomalies in nails. By streamlining these low-level features, RCBAM ensures the efficient processing of essential information while filtering out irrelevant details. As the network progresses to deeper layers, DHANet's advanced hybrid attention mechanism is introduced to extract high-level, nuanced patterns and context-aware relationships, for increased accuracy, as shown in studies that incorporate DHANet (Huang et al., 2024, Tsai et al., 2022). This will enable the model to analyze more complex features, such as subtle discoloration in skin or glossitis-related changes in the tongue, which require a more sophisticated understanding of interdependent visual cues. The combination of these two attention mechanisms—RCBAM for coarse attention refinement and DHANet for detailed context modeling—ensures high accuracy and accurate feature extraction across all modalities. All of this will be accessible through a mobile application, accessible to all with a smartphone and working camera (which is most of the global population (Howerth, 2024)). Therefore, our approach addresses the accessibility gap. Moreover, multimodal approaches like ours are predicted to achieve ~95% accuracy (Ramzan et al., 2024), significantly improving upon the unimodal methods that are more prevalent. This innovation holds promise for expanding equitable healthcare access to underserved populations, and detecting iron deficiencies before it becomes a life-threatening issue.

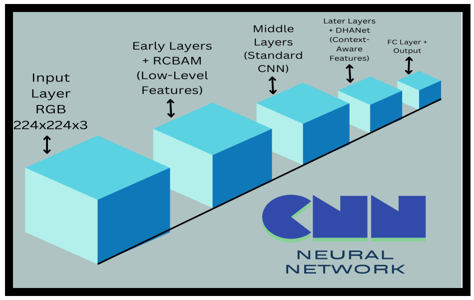
One of the primary components of the application is its ability to predict low iron levels using conjunctiva images. The conjunctiva, a highly vascularized tissue in the eye, provides a sensitive visual cue for early detection of hemoglobin levels (as shown in Nandi et al. and Zhao et al.) associated with iron deficiency, even before anemia develops. By analyzing subtle variations in the coloration and vascular patterns of the conjunctiva, the application will detect early signs of iron depletion through predicting hemoglobin levels (Chen et al., 2024). This will be achieved through a comparative analysis of the user's conjunctiva images against a dataset of labeled images from individuals with varying iron levels, enabling the model to identify patterns indicative of low iron. By focusing on these early changes, the application aims to address the root cause of anemia, promoting preventive action rather than reactive treatment.

While conjunctiva analysis plays a central role, integrating additional physiological indicators significantly strengthens diagnostic reliability and builds upon existing apps/technologies. We also aim for our application to incorporate images of nails, skin, and the tongue—other key markers of iron deficiency. For nails, the model analyzes features such as koilonychia (spoon-shaped nails), discoloration, and texture anomalies. Skin images are evaluated for pallor (paleness or yellow discoloration), particularly in the palms, while tongue analysis focuses on identifying discoloration, atrophic glossitis (when the tongue is smooth and lacks papillae, the small bumps), and other early signs of iron depletion (The Iron Clinic, 24). This unique, multimodal approach will enable our application to analyze a population database of individuals with known iron levels; the application will identify patterns (among predicted hemoglobin levels from conjunctiva images, and visual symptoms through nail, skin, and tongue images) that suggest low iron status. This integrative methodology not only aims to improve accuracy, but also enhances the model's ability to detect early-stage iron deficiency, providing users with actionable insights to prevent anemia from developing. Another factor that will set our model apart is its scalability and adaptability. It will be supported by a technology architecture backed by a robust data platform capable of incorporating incoming data from diverse populations across the globe. This infrastructure enables continuous updates to the AI model based on user feedback and new data inputs, ensuring the model grows more accurate and inclusive with time (Zhuravel, 2024). In turn, this ensures widespread applicability, facilitating equitable healthcare access for underserved populations while maintaining accurate diagnostic capabilities.

PROJECT LOGISTICS AND ORGANIZATION

The implementation of our project involves the creation of a classification CNN, in which two labels, 'iron deficient' and 'not iron deficient' are targeted by the model for classification. The input provided by the user would consist of RGB images taken directly from user mobile phones and plugged into our application's neural network. Our model, trained upon datasets containing images of physical features associated with iron deficiency like conjunctival eye pallor or the discoloration of the nail. Using a wide variety of datasets, including, but not limited to, the 218 RGB image collection Eyes-Defy-Anemia dataset as well as the human skin and fingernails images for non-invasive haemoglobin level assessment dataset, we hope to create a multimodal, reliable prediction model for iron deficiency detection. (Yakimov et al., 2024 and Fartale, 2024)

The architecture of our model is as follows: First, we have a CNN with an input layer that receives RGB images (e.g., 224×224×3) directly from a mobile phone camera. During the preprocessing stage of the images, we aim to use a dual approach to maintain the technical details of the image while still transforming it for our models' use. After rescaling the pixel values, we would apply mean and standard deviation normalization tailored to each dataset for better model performance. Afterwhich, we would utilize Contrast Limited Adaptive Histogram Equalization (CLAHE) to enhance image contrast, making it easier for our neural network to identify fine details in vascular patterns, discoloration, or textures (Marimuthu, 2024). In this way, we aim to provide our model with clean, transformed data that is then ready to be passed into the next layers of our CNN model for low-level feature refinement. Early layers of our CNN model would utilize RCBAM to assist in preserving original feature details while enhancing important feature characteristics through attention. By applying attention sequentially along channel and spatial dimensions, a refined feature map is pushed into the spatial attention model, resulting in a residual connection between the input feature map and output, creating emphasized data that allows for better prediction as the most important patterns are highlighted. (Zubair, 2024) The middle layers of our model would consist of Standard CNN Block which implements a convolutional layer to apply convolution filters to our input feature map to extract features and capture spatial patterns (e.g. textures and shapes) from the data. Then, after Batch Normalization to stabilize training and reduce model sensitivity and internal covariate shift, we would introduce non-linearity through the Rectified Linear Unit activation function. Finally, we would downsample the feature map to reduce spatial dimensionality using max pooling to make the model more efficient and dominant feature oriented. The later layers of our model would consist of a DHANet for high-level, context-aware feature extraction. This network is specifically designed to capture subtle patterns, such as discoloration, glossitis-related changes, and spatial relationships between features. DHANet uses hybrid attention mechanisms to ensure the model captures interdependencies between features as well as the global context, enabling the identification of subtle but meaningful features that are critical for accurate predictions. Finally, the fully connected layer outputs probabilities for the two classes, ‘iron deficient’ and ‘not iron deficient,’ using a softmax or sigmoid activation function, depending on the binary classification setup.



The following describes some risks and possible workarounds during implementation: If the model struggles with low accuracy, we will augment the dataset with diverse images from various demographics and lighting conditions. We will also fine-tune the CNN by adjusting hyperparameters like learning rate, batch size, and epochs. If all these measures fail, we can turn to data augmentation through generative adversarial networks like TimeGAN, where we create new data points based on the data we already have, assisting our models' predictive capability. If the model faces challenges in detecting subtle signs of iron deficiency, we will use data augmentation techniques to simulate a wider range of subtle variations in images (e.g. slight pallor in skin or minor changes in the conjunctiva), and may apply a multi-stage detection pipeline where initial low-confidence predictions are re-evaluated by a secondary model for more nuanced identification. If image quality is affected by camera resolution or lighting, we will develop an image preprocessing pipeline for correcting low light, shadows, and distortion, along with app guidelines for optimal image capture. A denoising autoencoder may also be used for clearer images.

We have already tested our idea with a standard CNN model and small datasets of iron deficiency visuals. Going forward, here are our milestones: **Implement the CNN model using RCBAM & DHANet:** Develop a CNN model with RCBAM and DHANet integrated to analyze images for iron deficiency prediction. We will evaluate the model's performance using accuracy, f_1 score, and MSE. We aim for an accuracy of over 85% at this time and an f_1 score greater than 0.75 in the initial test results. **Ensure multimodal approach works and begin app creation:** Integrate multimodal analysis using images from the conjunctiva, nails, tongue, and skin, and begin the app development process. The goal is to have a basic prototype where users can interact with the model and view results, a process we will design using Streamlit widgets. Afterwards, we will develop the final app using a native framework such as Swift for iOS or Kotlin for Android. **User testing and app release**: Conduct user testing with real-world data (in partnership with a local hospital, if applicable), collect feedback on app functionality, UI, and model accuracy, and finalize the app prototype. The app will be published on the Android and/or Apple App Store, and post-launch issues will be reviewed for potential updates. The aim is to ensure a user-friendly experience with an accuracy rate of over 95% at this point.

Timeline:

2/25/25 - Implement CNN model using RCBAM & DHANet. Document results and test model accuracy, f1_score, mean squared error, etc. → **3/10/25** - Ensure multimodal approach works as intended and begin app creation process. Record integration of multimodal data and the beginning stages of app UI development. → **3/29/25** - Continue coding app UI and integrating the classification CNN model, demonstrate prototype and evaluate validity of approach. → **5/01/25** - Finish app and begin user testing. If able, partner with a local hospital to test product on those with and without iron deficiencies. Document final app features and record user testing feedback, including issues found and improvements made. → **5/25/25** - Begin release process, working to publish the app online or within the Android/Apple app store.

We will evaluate our project as we go along. During the training stages of our model, we will continually use statistical analysis methods like the f1_score and accuracy score to check how our model performs. Once the model is completed to the best of our ability, we hope to partner with a local hospital to input actual patient data for our model to classify. Once we design our app, we plan to test the UI by allowing our family and friends to act as beta testers, and hopefully, eventually, a wider audience will be able to use our app when we roll it out on the app store. Generally, industry standards for machine learning model accuracy are 70% or above. However, since our neural network is meant to help identify iron deficiency and may lead to medical decisions based on its findings, we hope to reach an accuracy rate of 95% or higher using our multimodal approach.

We will facilitate collaboration through our in-person proximity, allowing for easy in-person meetings when necessary. Our code will be uploaded to GitHub to ensure seamless collaboration, enabling both of us to edit and update the project. We'll meet up to run and train the final model together, with access to the GPU.

Current Work: After reviewing prior studies on identifying iron deficiencies through visual cues, which revealed unimodal, inaccurate models not designed for public use, we were inspired to develop a multimodal identification model for an iron deficiency detection app accessible to everyday users. Our next step was to find the datasets that we'd use to train the model. After scouring the internet, we found around seven datasets we could use with features like nail discoloration indicating iron deficiency. We hope to grow this data pool. We then began to build our first, basic prototype model. At first, we attempted to use a KNN decision tree model, but found it lacked the nuance to identify milder symptoms in images. As such, we moved on to a CNN model. We tested a very standard CNN model with an input layer, convolutional layers, pooling layers, a fully connected layer and an output layer. We found that while our model had some more success in comparison to our initial KNN model, it still was unable to pick up on less obvious image patterns. This may be improved as we add more data sets to our training dataset, but we also are planning on adding attention mechanisms like RCBAM and DHANet to improve our accuracy.

Budget: We plan to allocate our budget for purchasing an external GPU and covering app publication costs. These expenses are essential for training our models efficiently and making the app accessible to users.

Item	Quantity	Cost	Link
GeForce RTX 4070 Super 12GB GDDR6X	1	$689.99	https://marketplace.nvidia.com/en-us/consumer/graphics-cards/asus-tuf-gaming-geforce-rtx-4070-super-12gb-gddr6x-oc-edition/
Apple App Store/ Google Play Publishing Cost	1	$223.00	https://www.spaceotechnologies.com/blog/cost-to-put-app-on-app-store/

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